Study 2: Longterm exercise and attention of young people (#48421)

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1) Have any data been collected for this study already?
No, no data have been collected for this study yet.

2) What's the main question being asked or hypothesis being tested in this study?
The aim of this study is to test which type of longterm (8 weeks) exercise intervention – strength-related, endurance-related or coordination-related – has the greatest benefit on the attention in a sample of people between 15 and 18 years old. We assume that the endurance-related intervention has a greater effect on attention than the strength-related and coordination-related interventions.

3) Describe the key dependent variable(s) specifying how they will be measured.
DV is the overall score of attention measured by the d2-R test of attention (Brickenkamp et al., 2010).

4) How many and which conditions will participants be assigned to?
8 conditions in a mixed pre-post design study: participants are randomly assigned to one of three intervention groups - endurance (HIIT) training, strength training, coordination training - or to a control group. All participants complete a pretest and a posttest, in which attention is measured.

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.
A 4x2 repeated measure ANOVA with four groups (endurance, strength, coordination, and control) and two measurements (pretest and posttest) will be conducted to investigate the change in attention from before to after the intervention.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.
We will compute the overall mean and standard deviation across all conditions, and winsorize at 3 SD above/below the mean.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.
An a priori sample-size calculation with G*Power for repeated measure ANOVA with four groups and two measurements, based on middle effect size ($f = 0.25$), power = 0.95 and $\alpha = .05$, resulted in a minimal sample size of 76 participants. Because we expect some drop-outs due to an 8-week study, we will start with 100 participants.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)
We will assess socio-economic factors and individual exercise history of the participants as potential covariates.